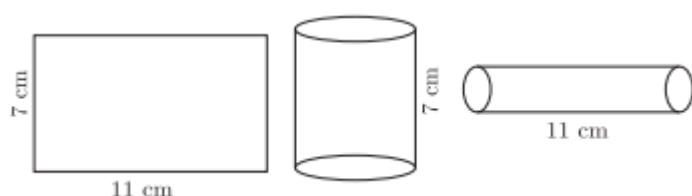


Chapter

Surface Area And Volume

1. A rectangular sheet of paper of size 11 cm # 7 cm is first rotated about the side 11 cm and then about the side 7 cm to form a cylinder, as shown in the diagram.

The ratio of their curved surface areas is : Main 2024



- (a) 1 : 1
- (b) 7 : 11
- (c) 11 : 7
- (d) $\frac{11\pi}{7} : \frac{7\pi}{11}$

2. Volume of a cylinder of height 3 cm is 48π . Radius of the cylinder is:
Main 2023

- (a) 48 cm
- (b) 16 cm
- (c) 4 cm
- (d) 24 cm

3. If two solid hemispheres of same base radius r are joined together along their bases, then curved surface area of this new solid is Comp 2023

- (a) $4\pi r^2$
- (b) $6\pi r^2$
- (c) $3\pi r^2$
- (d) $8\pi r^2$

4. The volume of a conical tent is 462 m^3 and the area of base is 154 m^2 . The height of the cone is : Main 2022 Sem II

- (a) 15 m
- (b) 12 m
- (c) 9 m
- (d) 24 m

5. The radius of a roller 100 cm long is 14 cm. The curved surface area of the roller is : Main 2022 Sem II

(Take $\pi = \frac{22}{7}$)

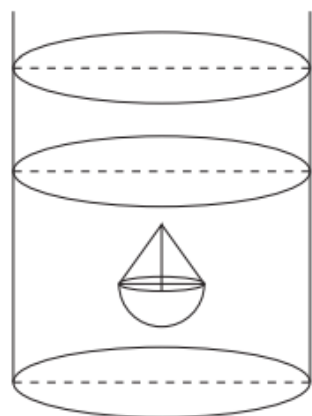
- (a) $13,200 \text{ cm}^2$
- (b) $15,400 \text{ cm}^2$
- (c) $4,400 \text{ cm}^2$
- (d) $8,800 \text{ cm}^2$

6. From a solid cylinder whose height is 8 cm and radius 6 cm, a conical cavity of same height and same base radius is hollowed out. Find the total surface area of the remaining solid. (Take $\pi = 3.14$.) Comp 2023

7. A solid cone of radius 5 cm and height 9 cm is melted and made into small cylinders of radius of 0.5 cm and height 1.5 cm. Find the number of cylinders so formed. Main 2022 Sem II

8. Small 126 cones, each of diameter 7 cm and height 3 cm, are melted and recast into a metallic sphere. Find the diameter of the sphere and hence find its surface area. Use $\pi = \frac{22}{7}$ Comp 2022

9. A solid is in the shape of a hemisphere of radius 7 cm, surmounted by a cone of height 4 cm. The solid is immersed completely in a cylindrical container filled with water to a certain height. If the radius of the cylinder is 14 cm, find the rise in the water level. Main 2023



Solution

1. **Given:** Rectangular sheet 11 cm×7 cm

Case 1: Rotated about side 11 cm

- Circumference of base = 11 cm
- Height = 7 cm

Curved surface area:

$$CSA_1 = (\text{circumference}) \times (\text{height}) = 11 \times 7 = 77$$

Case 2: Rotated about side 7 cm

- Circumference of base = 7 cm
- Height = 11 cm

Curved surface area:

$$CSA_2 = 7 \times 11 = 77$$

Ratio of curved surface areas:

$$CSA_1 : CSA_2 = 77 : 77 = 1:1$$

Correct Answer: (a) 1 : 1

2. **Given:**

Volume of cylinder = $48\pi \text{ cm}^3$

Height $h = 3$

Formula:

$$V = \pi r^2 h$$

$$48\pi = \pi r^2 \times 3$$

Cancel π :

$$48 = 3r^2$$

$$r^2 = 16 \Rightarrow r = 4 \text{ cm}$$

Correct Answer: (c) 4 cm

3.

- Curved surface area (CSA) of one hemisphere:

$$= 2\pi r^2$$

- For two hemispheres:

$$\text{Total CSA} = 2 \times 2\pi r^2 = 4\pi r^2$$

Correct Answer: (a) $4\pi r^2$

4. Given:

Volume of cone = 462 m^3

Area of base = 154 m^2

Formula:

$$\text{Volume of cone} = \frac{1}{3} \times (\text{area of base}) \times h$$

$$462 = \frac{1}{3} \times 154 \times h$$

Multiply both sides by 3:

$$1386 = 154h$$

$$h = \frac{1386}{154} = 9$$

Correct Answer: (c) 9 m

5. Given:

Radius $r = 14 \text{ cm}$

Length (height) $h = 100 \text{ cm}$

Take $\pi = \frac{22}{7}$

Curved Surface Area of a cylinder (roller):

$$\text{CSA} = 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 14 \times 100$$

$$= 2 \times 22 \times 2 \times 100 = 8800 \text{ cm}^2$$

Correct Answer: (d) $8,800 \text{ cm}^2$

6.

Given

Cylinder radius $r = 6$ cm

Height $h = 8$ cm

Conical cavity: same r and h

Take $\pi = 3.14$

Surfaces exposed

1. Curved surface of cylinder
2. Curved surface of cone (inside)
3. One circular base of cylinder

Calculations (board method)

1 Curved surface area of cylinder

$$2\pi rh = 2 \times 3.14 \times 6 \times 8 = 301.44$$

2 Curved surface area of cone

Slant height:

$$l = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$$

$$\pi rl = 3.14 \times 6 \times 10 = 188.40$$

3 Area of one circular base

$$\pi r^2 = 3.14 \times 36 = 113.04$$

✓ Total Surface Area

$$301.44 + 188.40 + 113.04 = \boxed{603.43 \text{ cm}^2}$$

(rounded as per board convention)

✓ Final Answer

$$\boxed{603.43 \text{ cm}^2}$$

7.

Given:

A solid cone is melted to form small cylinders.

Cone:

Radius $R = 5$ cm

Height $H = 9$ cm

$$\text{Volume of cone} = \frac{1}{3}\pi R^2 H = \frac{1}{3}\pi \times 25 \times 9 = 75\pi$$

One small cylinder:

Radius $r = 0.5$ cm

Height $h = 1.5$ cm

$$\text{Volume of one cylinder} = \pi r^2 h = \pi \times (0.5)^2 \times 1.5 = \pi \times 0.25 \times 1.5 = 0.375\pi$$

Number of cylinders:

$$\text{Number} = \frac{75\pi}{0.375\pi} = 200$$

✓ Answer:

200 cylinders

8.

Step 1: Volume of one cone

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 3$$

$$(3.5)^2 = 12.25$$

$$V_{\text{cone}} = \frac{22}{7} \times 12.25 = 38.5 \text{ cm}^3$$

Step 2: Volume of 126 cones

$$V_{\text{total}} = 126 \times 38.5 = 4851 \text{ cm}^3$$

Step 3: Volume of the sphere

$$\frac{4}{3}\pi R^3 = 4851$$

$$\frac{4}{3} \times \frac{22}{7} \times R^3 = 4851$$

$$R^3 = \frac{4851 \times 21}{88} = 1157.6$$

$$R \approx 10.5 \text{ cm}$$

Diameter of the sphere

$$\boxed{21 \text{ cm}}$$

Step 4: Surface area of the sphere

$$\text{Surface Area} = 4\pi R^2 = 4 \times \frac{22}{7} \times (10.5)^2$$

$$(10.5)^2 = 110.25$$

$$= 1386 \text{ cm}^2$$

 Final Answers

- Diameter of the sphere = $\boxed{21 \text{ cm}}$
- Surface area of the sphere = $\boxed{1386 \text{ cm}^2}$

9.

Step 1: Volume of the solid

(a) Volume of hemisphere

$$V_h = \frac{2}{3}\pi r^3 = \frac{2}{3}\pi(7)^3 = \frac{686}{3}\pi$$

(b) Volume of cone

$$V_c = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(7)^2(4) = \frac{196}{3}\pi$$

Total volume of solid

$$V = \left(\frac{686}{3} + \frac{196}{3} \right) \pi = \frac{882}{3}\pi = 294\pi \text{ cm}^3$$

Step 2: Rise in water level

Let rise in water level = h cm.

Volume displaced = volume of solid

$$\pi R^2 h = 294\pi$$

$$\pi(14)^2 h = 294\pi$$

Cancel π :

$$196h = 294$$

$$h = \frac{294}{196} = \boxed{1.5 \text{ cm}}$$

 **Answer**

$$\boxed{\text{Rise in water level} = 1.5 \text{ cm}}$$